

Answer: B

- 16** An interference pattern is formed on a screen by the superposition of two coherent light beams, one of which has four times the intensity of the other. The ratio of maximum to minimum intensity in the pattern is

A 1.7

B 3

C 4

D 9

Ans: D

If one of the light has four times the intensity of the other, it must have $\sqrt{4} = 2$ times the amplitude as $I \propto a^2$.

Therefore, constructive interference will yield 3 times the amplitude and destructive interference 1 time the amplitude.

As such, the ratio of the maximum to minimum intensity will be $3^2/1^2 = 9$ times.

- 17** If a transparent glass block is placed in front of one of the slits of a Young's double slit